



DATA WAREHOUSING

AND BIG QUERY



IN THIS PRESENTATION

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WHAT IS DATA WAREHOUSING?

- A Data Warehouse is a centralized repository for analytical data collected from multiple operational systems
 - Data warehouses answer business questions, not transactions
- Stores historical, integrated data
- Optimized for querying and reporting
- Supports business decision-making
- Some data warehouse vendors include:
 - Amazon Redshift
 - Azure Synapse Analytics
 - Google BigQuery

OLTP VS OLAP SYSTEMS

Online Transaction Processing (OLTP)

- Simple query type
- Many small transactions
- Insert, update, delete functions
- Normalized schema
- Uses current data
- For operations

Online Analytical Processing (OLAP)

- Complex queries
- Aggregations and joins
- Denormalized schema
- Uses historical data
- For insights

DATA WAREHOUSE VS DATA LAKE

Data Warehouse(DWH)

- Houses structured data
- Schema-on-write
- Business ready
- Good for reporting and dashboards

Data Lake

- Stores raw structured and unstructured data
 - Often object blobs or files
- Schema-on-read
- Flexible and low-cost storage
- Good for machine learning and AI workflows

DATA WAREHOUSE ARCHITECTURE

- Data Warehouses use a 3-tier architecture:
 - Bottom tier – Data sources and data storage
 - The core function of the bottom tier is the ETL process(Extract, Transform, Load)
 - Middle tier – OLAP engine processes complex analytical queries and bridges the bottom and top tiers
 - OLAP is a powerful technology for complex calculations, trend analysis, and data modeling
 - Top tier – Front end tools for interacting with the data stored and processed on the lower tiers
 - Includes business intelligence tools
 - Popular BI tools include IBM Cognos, Microsoft BI Platform, and Pentaho

OPERATIONAL DATA STORE

- An operational data store(ODS) sits between transactional systems and the data warehouse
- It is a type of database that stores integrated data from various sources within organizations
- It integrates data from multiple sources but stays up to date, unlike a data warehouse
- The ODS supports daily operations while the data warehouse supports analysis

DATA MART

- A Data Mart is a department-specific slice of a data warehouse built for targeted analysis
- We have two different types of data marts:
 - Dependent data mart – from the data warehouse
 - Independent data mart – direct from sources
- Most modern systems use dependent data marts to ensure consistency with enterprise data
- Data marts are often used by sales, finance, and marketing domains

DATA CLEANSING

- Data cleansing is the process of fixing or removing bad data
- It ensures data is accurate, complete, and consistent before analysis
- Removes duplicates, handles missing values, standardizes formatting, validates data ranges, and corrects inconsistencies
- Modern systems clean data continuously, not just once or twice
- Takes place during ingestion, in staging areas, and in DWH

WHAT IS BIG QUERY?

- Google BigQuery is a PaaS that supports querying using a dialect of SQL
- It is a fully managed, serverless, and AI ready data warehouse that supports both structured and unstructured data
- You can train and run machine learning models directly in SQL without exporting data

DENORMALIZATION

- Denormalization is an optimization process that introduces pre-computed redundancy into a normalized database
- It is a strategy for improving read performance of a database by adding or grouping data
- This reduces the need for complex joins
- It is applied after normalization to balance speed with data integrity
- Some common techniques include pre-joining tables, copied/mirrored tables, table splitting, and materialized views